



Association and Causality

APHS 20203 • Epidemiology & Biostatistics

Epidemiological Study: Fundamental Question

After identifying patterns (Person–Place–Time), we ask:

- **Are exposure and outcome linked?**
- Is the observed association valid or misleading?
- Could chance, bias, or confounding explain the finding?
- Does the evidence support a causal interpretation?

Disease Causality: Concepts Over Time

Causal explanations evolve with **available evidence, tools, and methods**.

- **Supernatural explanations:** Disease attributed to fate, spirits, or moral forces
- **Environmental (Miasma) theories:** Illness linked to air, water, climate, and surroundings
- **Germ theory:** Specific biological agents identified as causes of many infectious diseases
- **Risk-factor & multicausal models:** Most diseases arise from **multiple interacting factors**, not a single cause
- **Modern epidemiology:** Causal inference integrates biology, behavior, environment, and social context over time

What Do Epidemiologists Report First?

When studying health in populations, epidemiologists usually start by reporting:

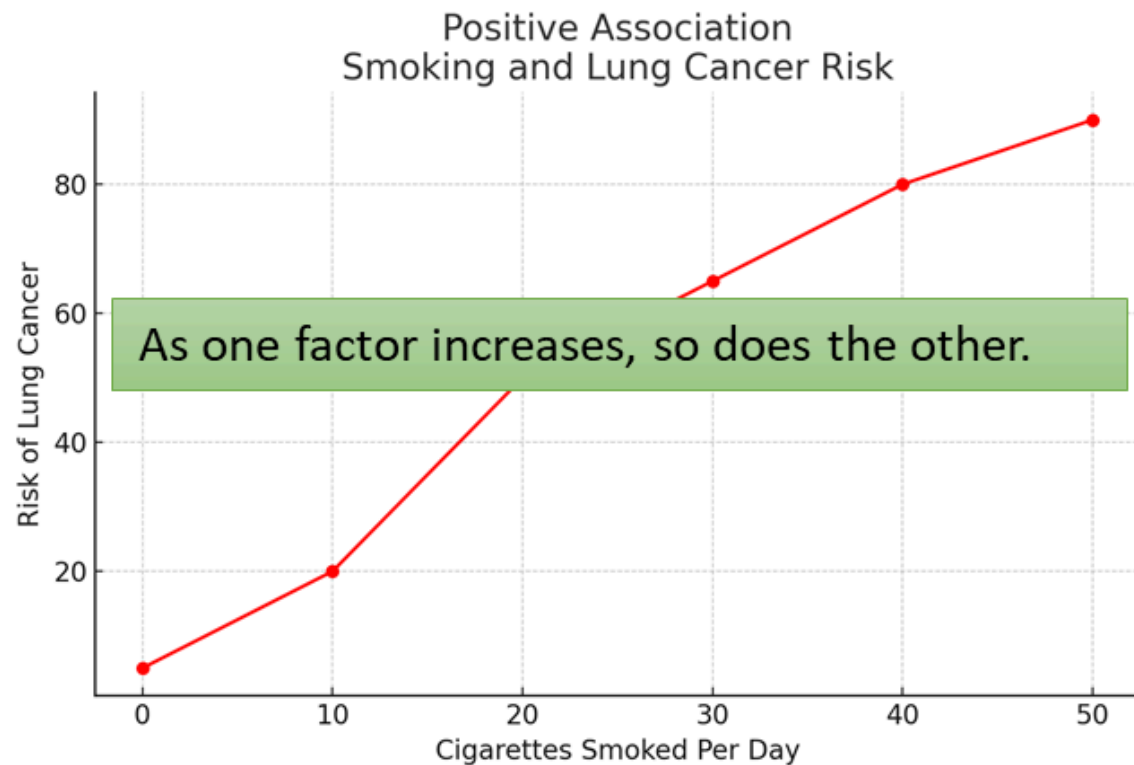
- A.** Clear cause-and-effect relationships
- B.** Patterns and correlations
- C.** Biological mechanisms
- D.** Random events with no structure

Association

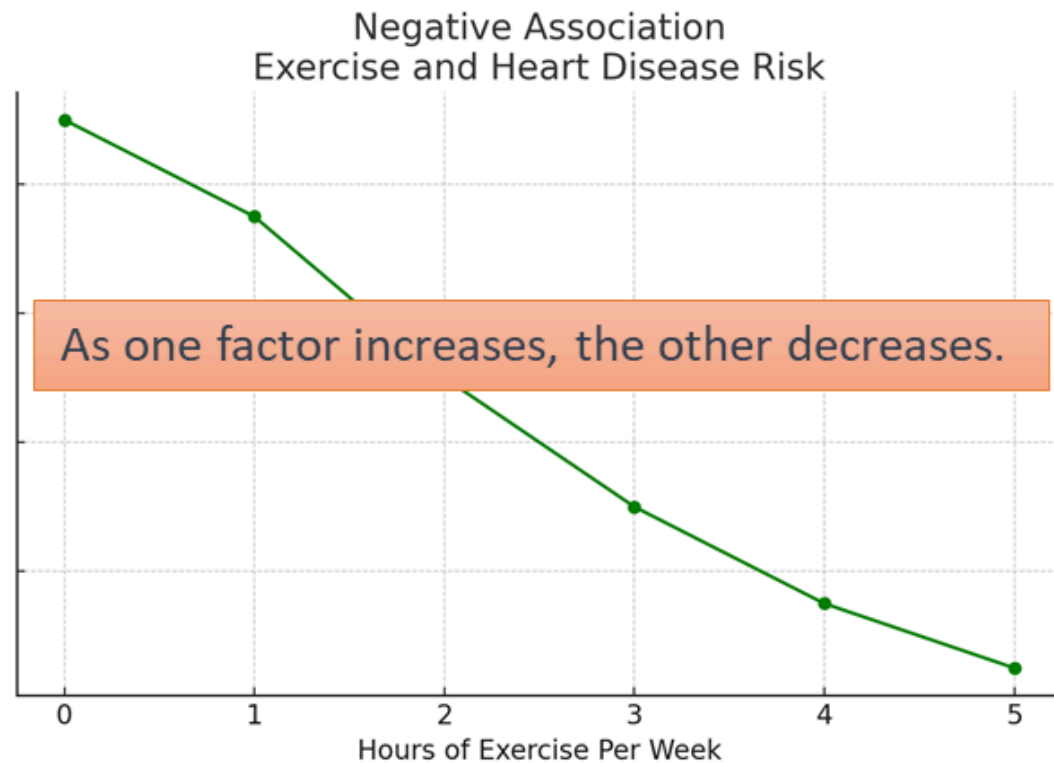
- **Association:** The distribution of an outcome differs across levels of an exposure.
 - For example, the outcome may be more common in one group than another.
- Knowing something about **X** helps us predict something about **Y**.
- **Types:** Positive association and negative association.
- An association alone does not tell us **why** the groups differ.

Types of Association

Positive Association



Negative Association



Causality

- **Causality:** A relationship where a change in one factor **directly produces** a change in an outcome.
- **Key distinctions**
 - **Causality implies association**
 - **Association does not imply causality**
- **Why causality is harder to establish**
 - Multiple contributing factors
 - Role of chance, bias, and confounding
 - Ethical and practical limits on experimentation

The Search for Causality

- When an association is observed, epidemiologists may ask:
 - **Is the association causal?**
Could chance, bias, or confounding explain it?
 - **Is the association consistent with what we (think we) know about how the world works?**
Across time, place, populations, and biology?
 - **Does the evidence support a causal interpretation?**
Or are alternative explanations more plausible?
- **Key idea:** Causality is not proven by a single study, but **evaluated through accumulated evidence.**

Bias

- **Bias:** *Systematic* error in an estimate. The gap between what we measured and what we intended to measure.
- Bias can arise from **who enters a study, how we measure variables, or how we estimate a causal effect.**
- **Example:** We want to estimate average physical activity among all Texas adults, so we measure physical activity patterns at local gyms.
- **Key idea:** A larger sample does not generally eliminate bias.

Confounding

- **Confounding:** The effects of other factors become mixed into our estimate of an exposure's causal effect. Or giving credit to the wrong variable.
- **Example:** Ice cream sales and drownings both increase during hot weather.
 - Hot weather influences both ice cream sales and drowning risk.
- **Question:** Would reducing ice cream sales reduce drownings?
- **Key idea:** The association can be real and useful for prediction without representing a causal effect of ice cream sales.

Exercise: Sunscreen Use and Skin Cancer

- **Observed association:** Higher rates of skin cancer are associated with increased sunscreen use.
- **Question:** What might **confound** the relationship between sunscreen use and skin cancer?
- **Think about:**
 - Who is more likely to use sunscreen?
 - What other exposure might be related to both sunscreen use and skin cancer?

Exercise: Which Is a Spurious Association?

Which scenario shows a **spurious association**?

- A.** As outside temperature drops, more people wear hats.
- B.** As vaccination rates increase, the targeted disease decreases.
- C.** Fires attended by more firefighters tend to cause more property damage.
- D.** People who exercise regularly have lower rates of heart disease.

Population, Sample & Uncertainty

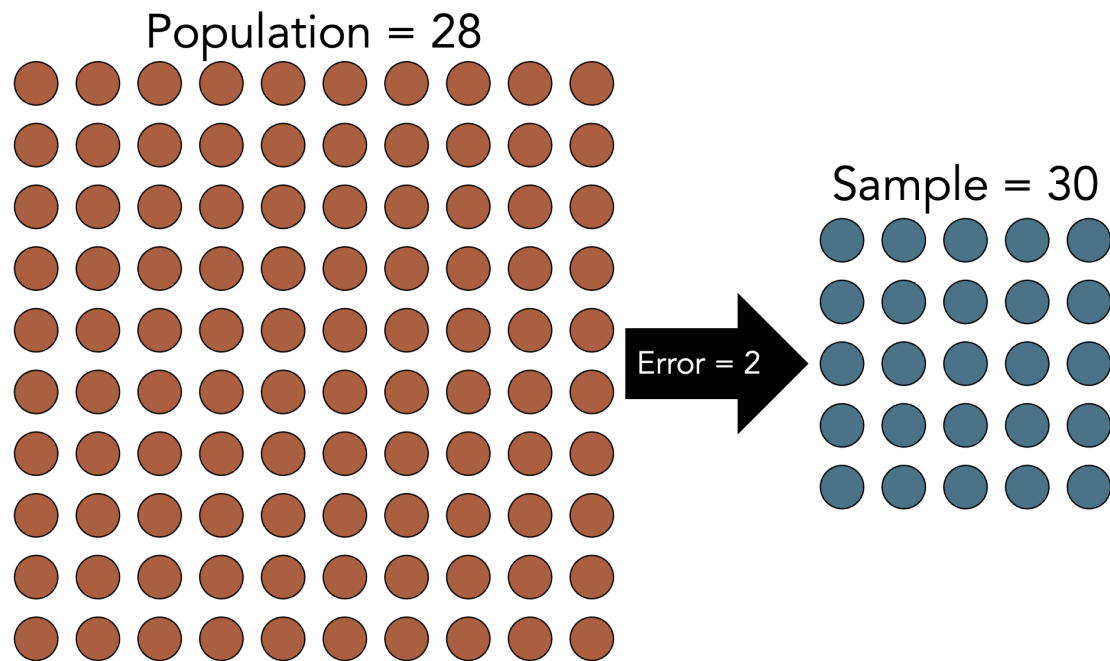
Example: We want to understand the health of adults living in Texas.

Group	Who belongs to it?	In our example
Target population	People we want findings to apply to	All Texas adults
Source population	People from whom we select participants	Adult patients at several Texas clinics
Study sample	People whose data we analyze	500 participating patients

- **Chance:** Would another random sample from the same source give a different estimate?
- **Applicability:** How well do these findings apply to all Texas adults?

Role of Chance in Association/Causality

Sampling variability: Estimates differ from sample to sample, even when we sample randomly from the same population.



Hypothetical BMI example

- Population mean: **28**
- One random sample: **30**
- Another random sample might have a mean of **26**.

Question: Must a difference between an estimate and the truth mean someone made a mistake?

Controlling for Chance in Epidemiologic Studies

- **Larger samples** generally reduce sampling variability.
- **Systematic bias is a different problem.** A larger sample does not typically correct biased recruitment, measurement, or confounding.
- **Precision and causality are different questions.** We can estimate an association precisely and still be uncertain whether it is causal.
- **Key idea:** More data can improve precision, but causal interpretation still requires careful judgment.

Types of Causality

- **Deterministic causality**

- A cause leads to an outcome **with certainty**
- If the cause occurs, the outcome follows
- *Everyday example:* Pressing a light switch → the light turns on

- **Probabilistic causality**

- A cause **changes the chance** of an outcome
- The outcome does not always occur
- *Everyday example:* Studying more → higher chance of doing well on an exam
- Epidemiology focuses mostly on **probabilistic causality**.

Deterministic Causality in Epidemiology (Rare)

- **Defining feature:** Exposure leads to disease **with near certainty**
- **Why it's unusual**
 - Most diseases involve **multiple interacting causes**
 - Individual susceptibility and context matter
- **Epidemiologic example**
 - Untreated rabies virus exposure → rabies disease
- **Key takeaway:** Deterministic causality exists in epidemiology, but it is **rare and exceptional**.

Probabilistic Causality in Epidemiology (Typical)

- **Defining feature:** Exposure **increases the likelihood** of disease
- **Why this is typical**
 - Diseases arise from **multiple interacting causes**
 - Susceptibility varies across individuals
- **Epidemiologic example**
 - Smoking → higher risk of lung cancer
- **Key takeaway:** Most epidemiologic causal relationships involve **risk, not certainty**.

Exercise: Which Best Describes Deterministic Causality?

Which example best fits **deterministic causality**?

- A.** A medication leads to different levels of improvement across patients
- B.** A specific bacterium always causes disease in people without immunity
- C.** Increased exercise in a population is linked to lower heart disease rates
- D.** Rising global temperature is correlated with rising sea levels

Multivariate Causality

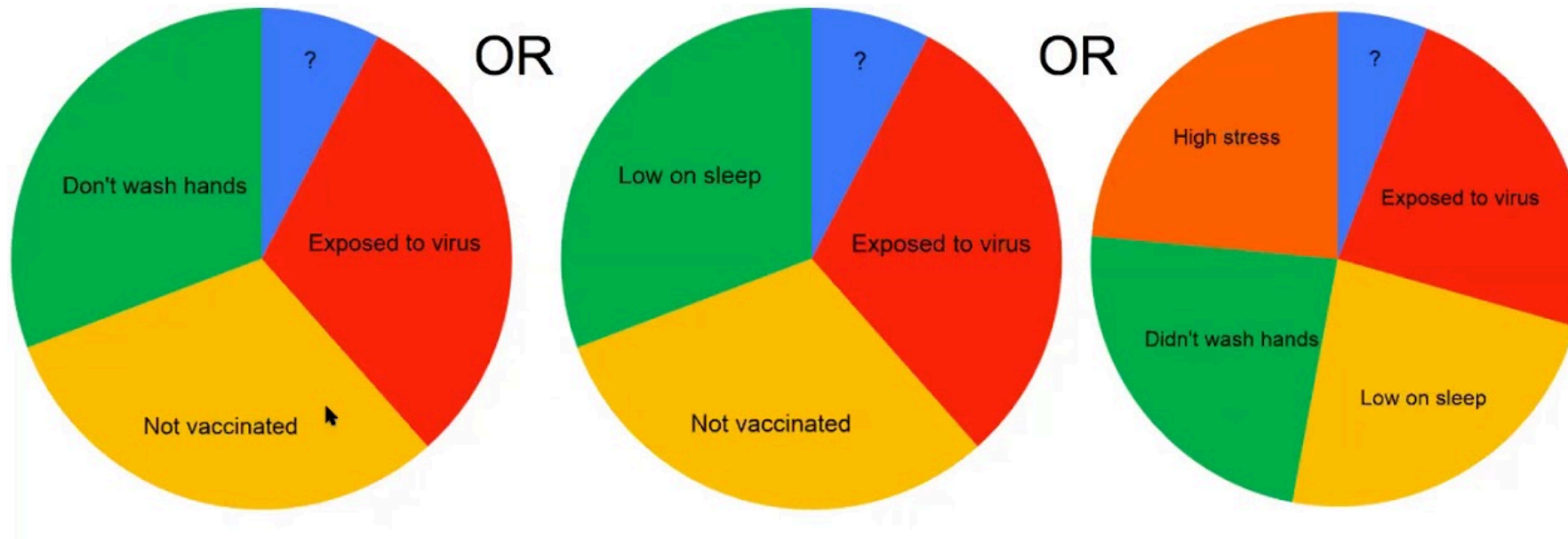
- Most health outcomes involve **multiple contributing causes**.
- Causes can act together, and different combinations can produce the same outcome.

Return to the light switch: What else must be present for flipping the switch to turn on the light?

- Electricity, a working bulb, and a complete circuit.
- Flipping the switch may complete the combination of conditions needed for the light to turn on.

Key idea: The final event in a sequence is not necessarily the only cause.

Rothman's Model of Sufficient and Component Causes (RMSCC)



Outcome: Flu

- Each pie is a **sufficient cause set** - "sufficient" to produce the outcome
- Each slice represents a **component cause**
- Different combinations of slices can produce the **same outcome**
- A **necessary cause** is a component cause that appears in every sufficient cause set

Project Prep 2

Quick Review: Association vs Causality

What distinguishes a **causal relationship** from an **association**?

- A.** A causal relationship means the two factors increase at the same time.
- B.** A causal relationship means changes in one factor lead to changes in another, with evidence that the exposure occurs before the outcome.
- C.** An association requires a statistically significant p-value, while a causal relationship does not.
- D.** Associations can only be observed in laboratory settings, while causal relationships are observed in real-world settings.

Hill's Criteria of Causality: Overview

- **Context:** Proposed by Sir Austin Bradford Hill (1965) to guide causal reasoning in epidemiology
- **Purpose:** A set of principles to help evaluate whether an observed **association** is likely to be **causal**
- **Important clarifications**
 - Not a checklist
 - Not all criteria must be met
 - **Temporality** (exposure precedes outcome) is essential
- **Key idea:** Hill's Criteria **organize evidence**; they do not prove causation.

Temporality & Strength of Association

Temporality

- The exposure must occur before the outcome.
- *Example:* Smoking precedes lung cancer development.
- *Counterexample:* A rooster crows before sunrise. Does it cause sunrise?

Strength of association

- A large association can support a causal interpretation, but alternative explanations still matter.
- *Counterexample:* Children's shoe sizes and reading aptitude are associated. What could explain both?

Key idea: Temporality is necessary, but neither timing nor strength alone establishes causality.

Consistency and Coherence

Consistency

- Similar findings across studies, populations, and settings.
- *Example:* High salt intake is linked to high blood pressure in many countries.
- **Limitation:** Studies that share the same bias can produce consistently misleading findings.

Coherence

- Findings fit with existing biological and epidemiologic knowledge.
- *Example:* Smoking-related lung cancer aligns with known carcinogenic effects of tobacco.

Comparison: Consistency compares studies. Coherence compares findings with existing knowledge.

Dose-Response Relationship & Biological Plausibility

- **Dose-Response Relationship (Biological Gradient)**

- Higher levels of exposure are associated with higher risk of the outcome
- A clear pattern strengthens confidence in a causal interpretation
- *Example:* Smoking more cigarettes per day is associated with higher lung cancer risk

- **Biological Plausibility**

- A reasonable biological mechanism links exposure to outcome
- The explanation should align with existing biological knowledge
- *Example:* Carcinogens in tobacco smoke damage lung cells and increase cancer risk

Experiment and Specificity

- **Experiment**

- Evidence comes from interventions or natural experiments
- Changes in exposure are followed by changes in outcomes
- *Example:* Smoking bans are followed by fewer asthma-related hospitalizations

- **Specificity**

- A specific exposure is linked to a specific outcome
- This criterion is less emphasized today because many exposures have multiple effects
- *Example:* Mycobacterium tuberculosis bacteria cause tuberculosis

Analogy (Weakest Criterion)

- **Analogy**

- Similar exposures are known to cause similar outcomes
- Raises a hypothesis by comparison, not by direct evidence
- *Example:* Other viruses are known to cause birth defects, raising concern about Zika virus

- **Why this criterion is weak**

- Relies on **similarity**, not direct observation
- Does not test exposure, timing, or mechanism
- Easily misleading if used alone

- **Key point:** Analogy can **suggest** a causal relationship, but it provides the **weakest support** and **cannot establish causality**.

Best Practices for Using Hill's Criteria

- **Start with temporality:** If exposure does not precede outcome, causality cannot be established
- **Evaluate criteria collectively:** No single criterion is sufficient. Temporality is necessary, but it is not sufficient.
- **Consider alternative explanations:** Always assess confounding, bias, and chance
- **Interpret findings in context:** Align evidence with biology and existing research
- **Avoid checklist thinking:** Hill's Criteria guide reasoning but do not prove causation

Example: Sugar and Type 2 Diabetes

- **Observed finding:** People who consume more sugar have higher rates of type 2 diabetes.
- **How should we interpret this?**
 - **Spurious association?** Could a third factor (e.g., overall diet, physical activity) explain both?
 - **Causal relationship?** Is this a real pattern that still needs stronger evidence? What additional evidence would be needed to support causality?

Exercise: Sugar and Diabetes

Task: Identify whether this association is causal. If so, specify the type.

Statement 1: High sugar intake directly increases the risk of type 2 diabetes by causing insulin resistance.

Causality (probabilistic): Proposes a cause-and-effect relationship in which high sugar intake increases diabetes risk through insulin resistance, but not with certainty.

Statement 2: Studies have found a correlation between sugar-sweetened beverage consumption and the incidence of diabetes.

Association - Positive: Describes a correlation between two variables (sugar-sweetened beverage consumption and diabetes incidence) without establishing a direct causal link.

Exercise: Sugar and Diabetes Continued

Task: Identify whether this association is causal. If so, specify the type.

Statement: Consuming sugar does not guarantee that an individual will develop diabetes, but it significantly increases the probability.

Causality - Probabilistic: While sugar consumption increases the risk of diabetes, it does not do so with certainty for every individual, given the role of other factors.

Statement: In regions where sugar consumption has decreased, the rates of diabetes have also shown a decline.

Association – Positive: This statement could be seen as suggesting a positive association (if interpreted as “less sugar, less diabetes”) but is more about observing a trend over time. The causality (if implied) would be probabilistic due to the lack of direct cause-and-effect evidence provided.

The Search for Causality: EVALI Outbreak

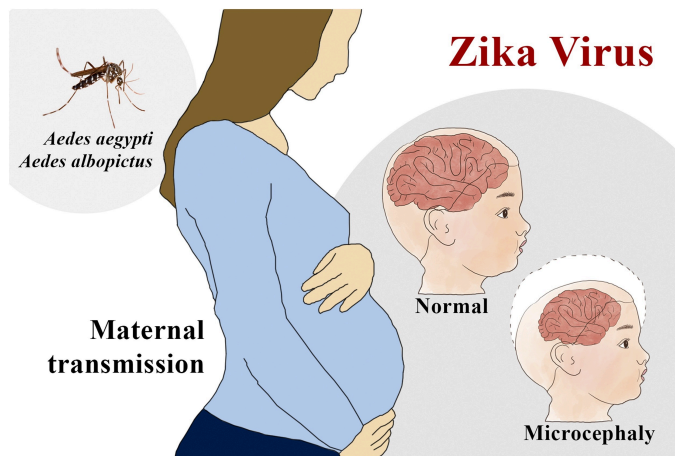
EVALI: E-cigarette or Vaping–Associated Lung Injury

Question: How did epidemiologists move from **observation** to **causal judgment**?

- **Observation (Early 2019):** Surge in lung injuries among people who reported vaping
- **Initial hypotheses:** Vaping products may be linked to lung injury: Which products? Which components?
- **Data collection (Mid-2019):** Patient histories, product testing, pattern comparison
- **Evidence accumulation (Late 2019):** Vitamin E acetate identified in many THC-containing products
- **Causal inference (2020):** Evidence supports a causal role for vitamin E acetate in EVALI

Example: Zika Virus & Microcephaly

- **Zika virus** is a member of the virus family Flaviviridae. It is spread by daytime-active *Aedes* mosquitoes.
- **Microcephaly** is a medical condition involving a smaller-than-normal head.



Atichat Kuadkitkan, Nitwara Wikan, Wannapa Sornjai, Duncan R. Smith, Zika virus and microcephaly in Southeast Asia: A cause for concern?, *Journal of Infection and Public Health*, Volume 13, Issue 1, 2020, Pages 11–15.

Group Activity: Zika Virus and Causal Reasoning

- You will watch a short video about the Zika virus and microcephaly.
- Each group has been assigned a **different lens** for evaluating causality at this Google link: <https://tinyurl.com/W5GroupActivity> or via the QR code
- **While watching the video:**
 - Focus only on your assigned lens and the last slide about dose-response
 - Capture brief bullet points, not full sentences
- **After the video, your group should report:**
 - One piece of evidence that supports causality
 - One limitation or uncertainty related to your lens
- **As a whole class, we will then discuss:**
 - Whether there is evidence of a dose–response relationship, and
 - Why some causal criteria may be harder to demonstrate than others

Video: Zika Virus & Microcephaly

<https://youtu.be/t-N9mryaVpY?si=RGQuPVqqhxsc4rxm>

Causality Criteria: Zika Virus & Microcephaly

- **Temporality:** Pregnant mothers got Zika before having babies with microcephaly.
- **Strength:** Big spike in microcephaly where Zika outbreaks occurred.
- **Consistency:** Similar rises reported in multiple regions (e.g., Brazil, French Polynesia).
- **Specificity:** Most increases appeared where Zika was active (though microcephaly has other causes).
- **Dose-Response (Biological Gradient):** More severe or earlier Zika infections may lead to higher risk of microcephaly.
- **Biological Plausibility:** Zika infects developing brain cells, a direct biological link to smaller head size.
- **Coherence:** Fits with existing knowledge that viruses can damage fetal brain development.
- **Experiment:** No direct clinical trials (ethical limits), but strong observational and lab evidence.
- **Analogy:** Similar viruses (e.g., rubella) cause birth defects, supporting the Zika-

Putting It All Together

- Causal inference in epidemiology requires careful judgment.
 - Associations are the starting point
 - Causality is evaluated using multiple lines of evidence
 - Most diseases have **multiple contributing causes**
 - Uncertainty and chance are always present
- **Final takeaway:** Epidemiology is about making the **best possible causal judgments** with imperfect evidence.